

This is the longest and most demanding laboratory in this course.

It covers a large amount of material and requires significant time to complete. Please plan accordingly and **do not leave it until the last moment**. This is a very important laboratory — complete all mandatory tasks carefully and thoroughly.

Submission deadline: 18.05.2026

During the class, we will follow a course that has been proposed by Google:

<https://developer.android.com/courses/android-basics-compose/course>

Tasks for the fourth lab:

1. Unit 4: Navigation and App Architecture

<https://developer.android.com/courses/android-basics-compose/unit-4>

2. Pathway 1: Architecture Components

<https://developer.android.com/courses/pathways/android-basics-compose-unit-4-pathway-1>

Learn about app architecture and how to use ViewModels, UI State, and StateFlow to build more complex apps.

a. Stages of the Activity lifecycle

<https://developer.android.com/codelabs/basic-android-kotlin-compose-activity-lifecycle>

b. ViewModel and State in Compose

<https://developer.android.com/codelabs/basic-android-kotlin-compose-viewmodel-and-state>

c. Write unit tests to test the ViewModel

<https://developer.android.com/codelabs/basic-android-kotlin-compose-test-viewmodel>

d. Practice: Add a ViewModel to Dessert Clicker (**mandatory**)

<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-viewmodel>

3. Pathway 2: Navigation in Jetpack Compose

<https://developer.android.com/courses/pathways/android-basics-compose-unit-4-pathway-2>

Learn how to use the Navigation component to build complex apps with multiple screens.

a. Navigate between screens with Compose

<https://developer.android.com/codelabs/basic-android-kotlin->

[compose-navigation](#)

- b. Test the Cupcake app
<https://developer.android.com/codelabs/basic-android-kotlin-compose-test-cupcake>
- c. Practice: Add navigation (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-navigation>

4. Pathway 3: Adapt for Different Screen Sizes

<https://developer.android.com/courses/pathways/android-basics-compose-unit-4-pathway-3>

Learn how to use window size classes, canonical layouts, and other techniques to build adaptive UIs optimized for all form factors.

- a. Build an adaptive app with dynamic navigation
<https://developer.android.com/codelabs/basic-android-kotlin-compose-adaptive-navigation-for-large-screens>
- b. Build an app with an adaptive layout
<https://developer.android.com/codelabs/basic-android-kotlin-compose-adaptive-content-for-large-screens>
- c. Practice: Build Sports app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-sports-app>
- d. Project: Create a My City app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-my-city>

As a result of the work please take the Quiz at the end of each Pathway.

Submission requirements:

- Use the **private** repository named MAD created during Lab 2.
- Inside the repository maintain the following folder structure:

```
MAD/  
  UNIT4/  
    DessertClicker      (Practice: Add a ViewModel to Dessert Clicker)  
    Navigation          (Practice: Add navigation)  
    SportsApp           (Practice: Build Sports app)  
    MyCityApp/          (Project: Create a My City app)
```

- The instructor should already be a collaborator (rmaciaszczyk@zut.edu.pl). If not, add them using that email address.

- Submit the repository link as a response to this assignment in **Microsoft Teams**.